STATUS AND DISTRIBUTION OF FRESHWATER MUSSELS (BIVALVIA: UNIONACEA) INHABITING LITTLE RIVER FROM THE ARKANSAS – OKLAHOMA STATE LINE TO MILLWOOD LAKE, ARKANSAS.

Chris L. Davidson[1], Tony Brady[2], Timothy Fotinos[3]

[1]U.S. Fish and Wildlife Service, Arkansas Ecological Services Field Office, Conway, AR 72032

[2]U.S. Fish and Wildlife Service, Natchitoches National Fish Hatchery, Natchitoches, LA 71457

[3]U.S. Fish and Wildlife Service, Red River National Wildlife Refuge, Bossier City, LA 71112



February 26, 2014

INTRODUCTION

The Little River in southeastern Oklahoma and southwestern Arkansas has a diverse mussel assemblage including the federally protected Ouachita Rock Pocketbook (*Arkansia wheeleri*), Rabbitsfoot (*Quadrula cylindrica cylindrica*), winged mapleleaf (*Quadrula fragosa*), and pink mucket (*Lampsilis abrupta*). Several researchers over the past several decades have sampled mussels in much of the river (Ecosearch 1987; Harris and Gordon 1987; Galbraith *et al*. 2005, Vaughn 1994; Vaughn *et al*. 1995; Vaughn and Taylor 1999; Seagraves 2006; URS 2007; Arkansas Game and Fish Commission (AGFC) Mussel Database 2014). Mussel declines in the Little River have largely been attributed to impoundments (Vaughn and Taylor 1999) and degraded water quality from point source effluents (Ecosearch 1987). The factors limiting recruitment also include threats affecting their fish hosts (Haag and Warren 1997; Vaughn and Taylor 2000).

Extensive systematic inventories of the Little River mussel fauna have been conducted from Pine Creek Reservoir to the Arkansas – Oklahoma state line (Vaughn 1994; Vaughn *et al.* 1995; Vaughn and Taylor 1999). A comprehensive survey from the Arkansas – Oklahoma state line to Millwood Lake is still needed to better understand the river's diverse mussel fauna. We spatially mapped mussel beds and assessed distribution and relative abundance for this reach of the Little River. We are providing this information so future conservation efforts, including land management activities at Pond Creek National Wildlife Refuge, can better protect the mussel fauna.

HISTORICAL LITTLE RIVER MUSSEL SURVEYS

Prior to the early 1990s, mussel surveys of the Little River in Oklahoma downstream of Pine Creek Reservoir were limited to sites with pedestrian access. Early (prior to 1983) survey efforts of the Little River from the Arkansas – Oklahoma state line to Millwood Lake also were limited to a few sites with pedestrian access. The entire Little River from Pine Creek Reservoir to the Arkansas – Oklahoma state line was surveyed in the 1990s (Vaughn 1994; Vaughn *et al.* 1995; Vaughn and Taylor 1999). From 1983 – 2008, survey efforts between the Arkansas – Oklahoma state line and Millwood Lake were limited to surveyor selected sites based on visible evidence of shell material or efforts to locate Ouachita Rock Pocketbook (Clarke 1987; Harris and Gordon 1987; Vaughn et al. 1995; AGFC Mussel Database 2014). These latter survey efforts are discussed in more detail below and summarized in Table 1. Mussel species (n = 44) historically documented from the Little River, Oklahoma and Arkansas, are listed in Table 2.

Pine Creek Reservoir, Oklahoma, to Arkansas-Oklahoma State Line

Ecosearch (1987) surveyed six sites in the Little River from near Millerton to the Arkansas – Oklahoma state line, McCurtain County, Oklahoma. Ecosearch's qualitative search efforts were focused on locating Ouachita Rock Pocketbook. Fourteen (14) species were identified at these sites.

From 1992 – 1993, Vaughn (1994) surveyed 17 sites from Pine Creek Reservoir to U.S. Highway 259/70 near Idabel, McCurtain County, Oklahoma. Twenty-seven (27) species were identified among these

sites. This represents the first attempt to conduct a comprehensive mussel inventory via boat using dive and snorkel techniques. Mussel surveys were timed to standardize sampling effort. Quadrat sampling techniques also were used at some sites prior to timed-searches (Vaughn 1994).

In 1994, Vaughn *et al*. (1995) surveyed the remainder of the Oklahoma reach of the Little River not surveyed by Vaughn (1994), U.S. Highway 259/70 to the Arkansas – Oklahoma state line. Thirty-one (31) species were identified among 16 sites located in Oklahoma. Their survey also included approximately five miles in Arkansas extending from the state line to the Rolling Fork Creek confluence. Eighteen (18) species were identified at five Arkansas sites.

Vaughn and Taylor (1999) examined the distribution and abundance of mussel s at 37 sites along a 240 km length of the Little River in Oklahoma. Thirty-two (32) of these sites were located downstream of Pine Creek Reservoir. Thirty (30) species were identified among these 32 sites. Vaughn and Taylor (1999) found with increasing distance from Pine Creek Reservoir, there was a gradual, linear increase in mussel species richness and abundance. Rare species only occurred at sites furthest from the reservoir. These same trends, greatly reduced mussel abundance, were apparent downstream of the Mountain Fork, a major tributary, inflow. In both situations, reservoir inflows to the Little River greatly reduced abundance of common, widespread species. While no species extirpations are known from the Little River, the biological integrity of numerous subpopulations has been greatly decreased by the loss of individuals (Vaughn and Taylor 1999).

In 2005, Galbraith *et al*. (2005) sampled five known mussel beds in the Little River between Idabel, Oklahoma, and the Mountain Fork confluence. Sample methodology included quantitative (quadrats) and semi-quantitative (timed searches) methods. Twenty-four (24) species were identified among the five sites.

Arkansas – Oklahoma State Line to Millwood Lake

Gordon *et al*. (1980) summarized the Arkansas distribution of mussels based on museum collections, and the historical mussel surveys that included Call (1895), Meek and Clark (1912), Wheeler (1914, 1918), and Hinkley (1916). In 1983, Harris and Gordon (1987) initiated surveys in search of Arkansas' rare mussels including Ouachita Rock Pocketbook. They surveyed the entire Little River from the Arkansas – Oklahoma state line to Millwood Lake, selecting nine sites for inventory based on visible evidence of shells. Twenty-eight (28) species were identified at these sites.

Ecosearch (1987) surveyed six sites in the Little River from near the Arkansas – Oklahoma state line to near Alleene, Little River County, Arkansas. Ecosearch's qualitative search efforts were focused on locating Ouachita Rock Pocketbook. Eighteen (18) species were identified among these sites.

In 2002, AGFC selected seven sites for inventory. The focus of this survey effort was to locate Ouachita Rock Pocketbook (Posey, pers. comm.). Twelve (12) species were identified at these sites (AGFC Mussel Database 2014).

Table 1. Summary of Little River (Pine Creek Reservoir to confluence with Red River) mussel survey data (1979 – 2013). NA = not applicable.

Little River	Source	Stream Miles (kilometers)	Sites	Sample Method	Number of Taxa
Pine Creek Reservoir to Arkansas – Oklahoma state line	Ecosearch (1987)	Pedestrian access	6	Qualitative	14
	Vaughn (1994)	25 (40.2)	17	Qualitative (time search) and quantitative	27
	Vaughn *et al*. (1995)	31.5 (50.6)	16	Qualitative (time search) and quantitative	31
	Vaughn and Taylor (1999)	40 (64.3)	32	Qualitative (time search) and quantitative	30
	Galbraith *et al.* (2005)	NA	5	Qualitative (time search) and quantitative	24
Arkansas – Oklahoma state line to Millwood Lake	Harris 1979 in AGFC Mussel Database	NA	1	Qualitative	16
	Robison 1978 in AGFC Mussel Database	NA	1	Qualitative	12
	Ecosearch 1987	Pedestrian Access	6	Qualitative	18
	Harris and Gordon (1987)	Surveyor selected sites based on visible evidence of shells	9	Qualitative	28
	Vaughn *et al*. (1995)	5.5 (8.8)	4	Qualitative (time search)	18
	AGFC 2002 in AGFC Mussel Database	Surveyor selected sites	7	Qualitative	12
	AGFC 2006 in AGFC Mussel Database	Surveyor selected sites	5	Qualitative	25
	AGFC & FWS	Surveyor selected sites based on visible evidence of shells	14	Qualitative; one site quantitative	28
Millwood Lake to Red River	Seagraves (2006)	Surveyor selected sites	18	Qualitative; one site quantitative	24
	AGFC 2005 in AGFC Mussel Database	Surveyor selected sites	1	Qualitative	6
	URS (2007)	1.3 (2.1)	54	Qualitative	21

Table 2. Mussel species (n = 44) historically documented from the Little River, Oklahoma and Arkansas.

Species	**Common Name**
Actinonaias ligamentina	Mucket
Amblema plicata	Threeridge
Arcidens confragosus	Rock Pocketbook
Arkansia wheeleri	Ouachita Rock Pocketbook
Ellipsaria lineolata	Butterfly
Elliptio dilatata	Spike
Fusconaia ebena	Ebonyshell
Fusconaia flava	Wabash Pigtoe
Lampsilis abrupta	Pink Mucket
Lampsilis cardium	Plain pocketbook
Lampsilis siliquoidea	Fatmucket
Lampsilis sp. B cf. *hydiana*	Louisiana Fatmucket
Lampsilis teres	Yellow Sandshell
Lasmigona complanata	White Heelsplitter
Lasmigona costata	Flutedshell
Leptodea fragilis	Fragile Papershell
Ligumia subrostrata	Pond Mussel
Megalonaias nervosa	Washboard
Obliquaria reflexa	Threehorn Wartyback
Obovaria jacksoniana	Southern Hickorynut
Plectomerus dombeyanus	Bankclimber
Pleurobema sintoxia	Round Pigtoe
Pleurobema rubrum	Pyramid Pigtoe
Potamilus purpuratus	Bleufer
Potamilus ohiensis	Pink Heelsplitter
Ptychobranchus occidentalis	Ouachita Kidneyshell
Pyganodon grandis	Giant Floater
Quadrula apiculata	Southern Hickorynut
Quadrula fragosa	Winged Mapleleaf
Quadrula cylindrica cylindrica	Rabbitsfoot
Quadrula nodulata	Wartyback
Quadrula metanevra	Monkeyface
Quadrula quadrula	Mapleleaf
Quadrula nobilis	Gulf Mapleleaf
Quadrula verrucosa	Pistolgrip
Strophitus undulatus	Squawfoot
Toxolasma parvus	Lilliput
Toxolasma texasensis	Texas Lilliput
Truncilla donaciformis	Fawnsfoot
Truncilla truncata	Deertoe
Uniomerus tetralasmus	Pondhorn
Utterbackia imbecillis	Paper Floater
Villosa iris	Rainbow
Villosa lienosa	Little Spectaclecase

The AGFC and U.S. Fish and Wildlife Service (Service) surveyed 14 sites between the Arkansas – Oklahoma state line and Millwood Lake during 2006 and 2008. Number of species per site ranged from seven to 22 with a mean of 14.7 (AGFC Mussel Database 2014).

Millwood Lake to Confluence with Red River

In 2002, Seagraves (2006) conducted the first survey of the Little River downstream of Millwood Lake. This inventory was focused on locating Ouachita Rock Pocketbook. Seagraves (2006) surveyed 18 sites in the 27 km reach and identified 23 species. Quantitative sampling of one mussel bed occupying 1,000 m^2 resulted in collection of 167 individuals representing 15 species. The mussel numerical standing crop estimate was 16,700 ± 3,916. Numerous species (*e.g.*, rabbitsfoot and winged mapleleaf) present upstream of Millwood Lake appear to be extirpated from the reach downstream of the lake. Abundance of common species also is reduced relative to upstream reaches.

URS (2007) surveyed the lower 2.1 km (54 transects) of the Little River. No mussel beds were identified within this river reach. Twenty-one (21) species were identified within the study area.

METHODS

Study Area

The Little River originates near Ludlow in LeFlore County, Oklahoma, and flows in a southwesterly direction before entering Pine Creek Reservoir, one of two main stem impoundments. Upon exiting Pine Creek Reservoir, the river flows in a southeasterly direction and enters Millwood Lake, approximately 56 km east of the Arkansas – Oklahoma state line, before converging with the Red River near Fulton in Hempstead County, Arkansas. The watershed drains an area of approximately 10,720 km^2 (Vaugn and Taylor 1999; Matthews *et al*. 2005). The headwaters and major tributaries are contained in the Ouachita Mountains and South Central Plains ecoregions, with the lower main stem wholly contained within the South Central Plains ecoregion (Woods *et al*. 2004). Matthews *et al.* (2005) discuss physiography, climate, and land use in the watershed and geomorphology, hydrology, chemistry, biodiversity, and ecology of the Little River.

Three impoundments influence the Little River (Vaughn and Taylor 1999). The main stem of the river is impounded by the 1,644 km^2 Pine Creek Reservoir, which began operation in 1969, and the 118 km^2 Millwood Lake, which began operation in 1966. The Mountain Fork, a major tributary of the Little River, is impounded by the 1,952 km^2 Broken Bow Reservoir, which was formed in 1968. Outflow from Broken Bow Reservoir enters the Little River 64 km downstream of Pine Creek dam (Vaughn and Taylor 1999).

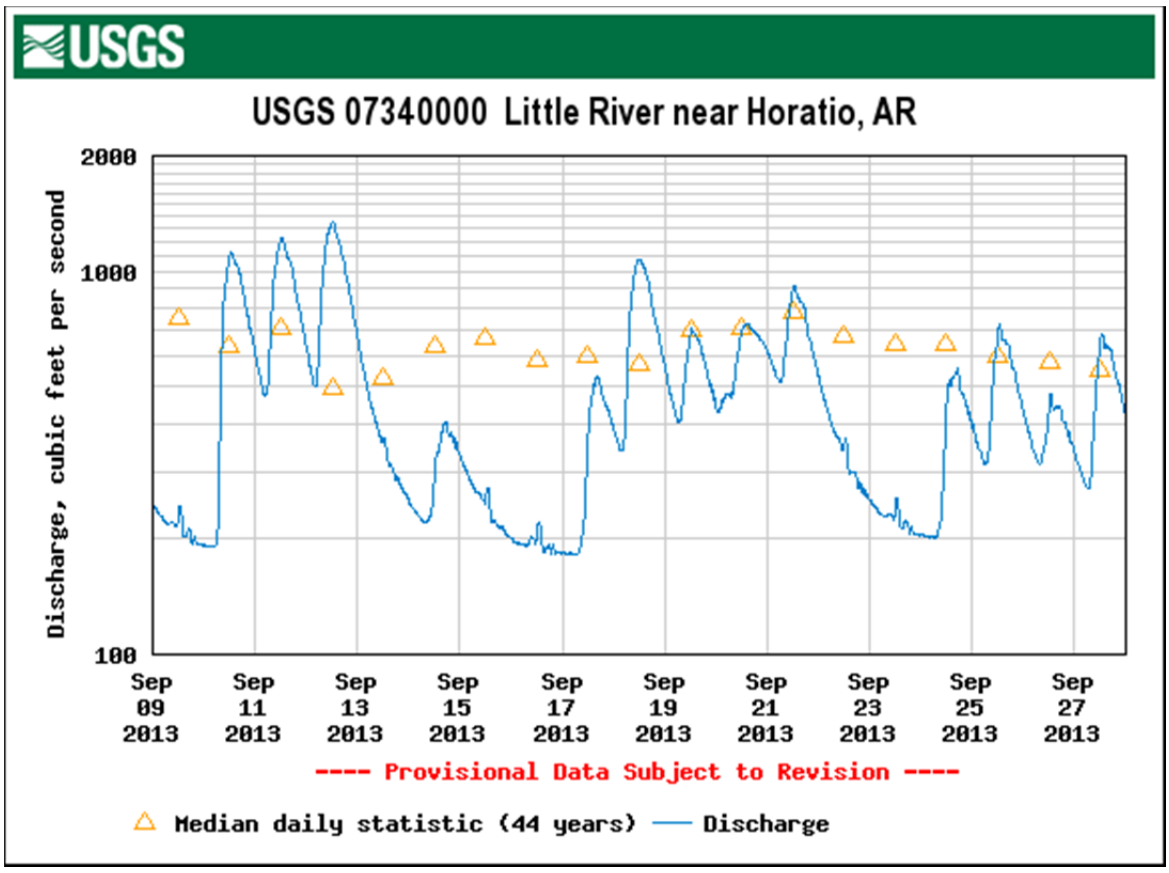
Pond Creek National Wildlife Refuge includes a portion of the Little River within and along its southern boundary. The refuge boundary begins approximately 1.3 km downstream of the Arkansas Highway 41 Bridge and extends to Millwood Lake. The refuge boundary is interrupted by five “inholdings” or private tracts adjacent to the river. Refuge management expressed a strong interest in a formal mussel survey

in the Little River due to the imperiled status and general lack of knowledge about the mussel community beyond searches for Ouachita Rock Pocketbook.

Survey Approach

This inventory project delineated existing mussel beds following sample methodology of Christian and Harris (2005) and provided initial estimates of species composition and relative abundance for mussels in the Little River between the Arkansas – Oklahoma state line and Millwood Lake. Survey sites were accessed via boat for the entire length of the Little River from the Arkansas – Oklahoma state line to the inundation pool of Millwood Lake. Each survey site was sampled using dive techniques. Each survey site was searched by two divers and search strategies included hand grubbing. Surveys were conducted from September 9 – 13 and 23 – 26, 2013. Discharge (flow) was generally less than median discharge for the 44 year period of record during this time frame (Figure 1). Nomenclature follows Turgeon *et al*. (1998).

Figure 1. Little River discharge (flow) near Horatio, Arkansas, from September 9 – 27, 2013 (USGS provisional data, http://waterdata.usgs.gov/ar/nwis/uv?site_no=07340000).



Initial transect searches of probable mussel habitats were from upstream to downstream to identify and determine the limits of a potential mussel bed. Transect searches generally consisted of an area extending bank to bank (*i.e.*, wetted channel width) and approximately 60 m in length. Divers estimated the number of live mussels per m^2 by tactile methods. If the number of mussels was ≥ 10/m^2, the diver estimated the width of the bed by traversing the limits in one meter increments. Additional downstream and upstream "transects" were conducted until the substrate was uninhabited by mussels or density was < 10/m^2. Total length of a bed was measured using a laser range finder. Water depth, determined by a depth finder, and substrate type were recorded. Relative abundance, species composition, and general habitat information were recorded for areas with mussel densities averaging < 10/m^2, but with no limit as to total area.

Population estimates and community numeric standing crop data have not been previously reported for the reach surveyed during our study. One mussel bed was quantitatively sampled in November, 2008, using methods of Harris *et al.* (1993), Christian (1995), and Christian and Harris (2005), the data from which is reported with the current study. Site 13-27 was selected due to the collection of a fresh dead Ouachita rock pocketbook (*Arkansia wheeleri*) located at the site during qualitative surveys conducted by the Service and AGFC in July, 2008 (C. Davidson, pers. comm.). Thirty-seven (37) quadrat sites were obtained from a random numbers table and applied in an X, Y coordinate style. Mussels within the quadrat were collected by excavating the substrate and visually or tactily searching through the substrate. Mussels were placed in a mesh bag and taken to the surface where they were identified, counted, and returned to the site of collection. We calculated species population estimates and assemblage total community numeric standing crop using the methods from Sampford (1962).

RESULTS

Survey of the 56 km of the Little River required 18 person days to complete. A total of 28 mussel beds were located and delineated (Table 3, Figures 2a – 2i). The mean bed area was 5,404 m^2 with a range of 240 to 15,525 m^2. Thirty-four (34) sites were searched outside of defined mussel beds (Table 4, Figures 2a – 2i).

A total of 4,097 individuals and 32 species were recorded during the qualitative surveys (Table 5). Mean species richness for qualitative surveys was 10.1 with a range of 0 – 22 (Table 5). Four species Pimpleback (*Quadrula pustulosa*), Mucket (*Actinonaias ligamentina*), Threehorn Wartyback (*Obliquaria reflexa*), and Deertoe (*Truncilla truncata*), comprised 77 percent of the total. Three federally protected species, Ouachita Rock Pocketbook (*Arkansia wheeleri*), Winged Mapleleaf (*Quadrula fragosa*), and Rabbitsfoot (*Quadrula cylindrica cylindrica*), were collected within the study area. One live and one dead Ouachita Rock Pocketbook and Winged Mapleleaf were collected from four previously undocumented sites. Rabbitsfoot was rarely abundant but relatively common throughout the study area. Mean length of Rabbitsfoot was 96.4 mm with a range of 56.0 – 116.1 mm (Figure 3). Mean number of live individuals sampled per site for all species was 67.2 with a range of 0 – 374 (Table 5). Catch per unit effort (CPUE) per site averaged 1.4 individuals/minute and ranged from 0 – 6.0 individuals/minute.

Figure 2a. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

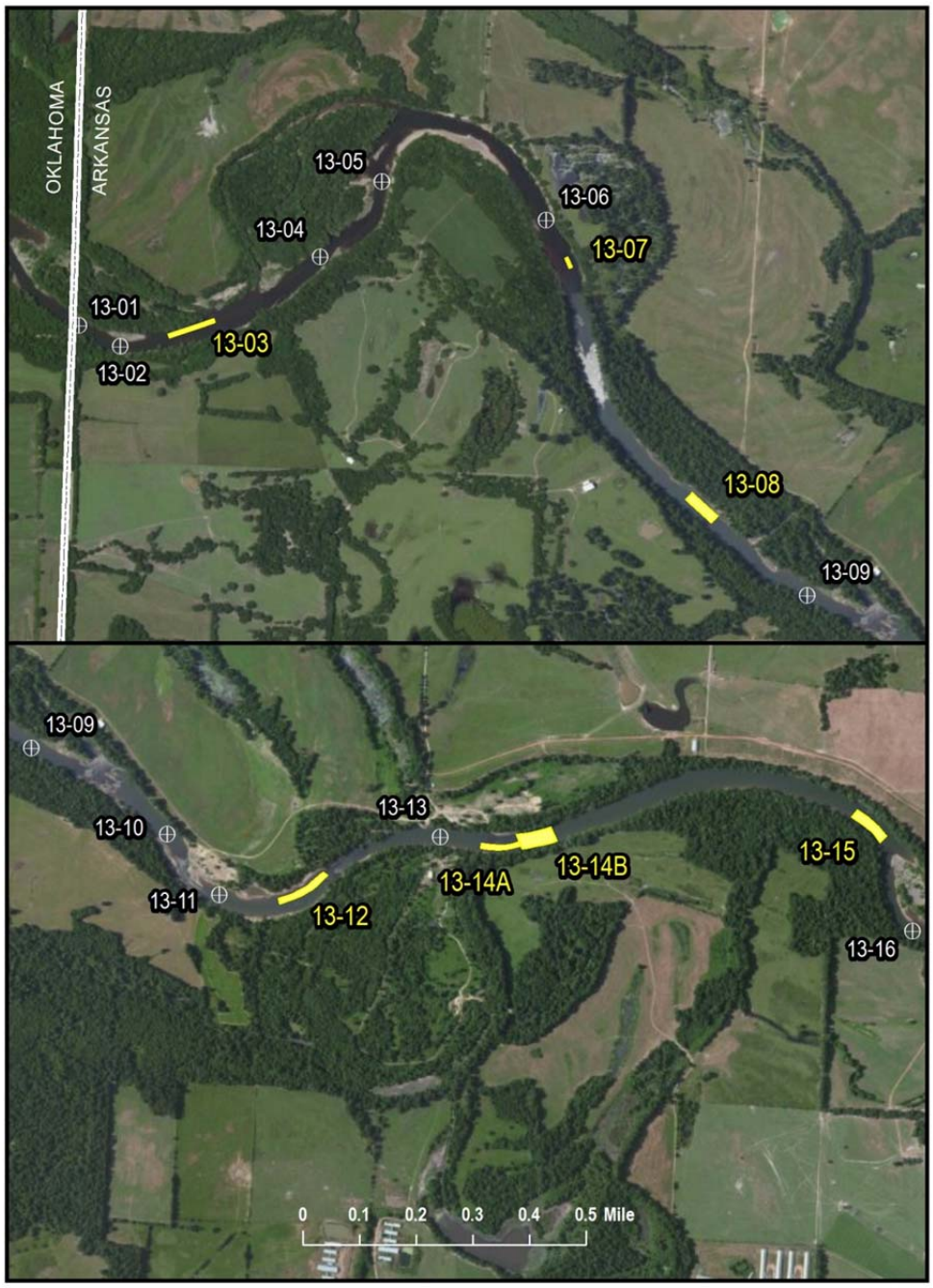


Figure 2b. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

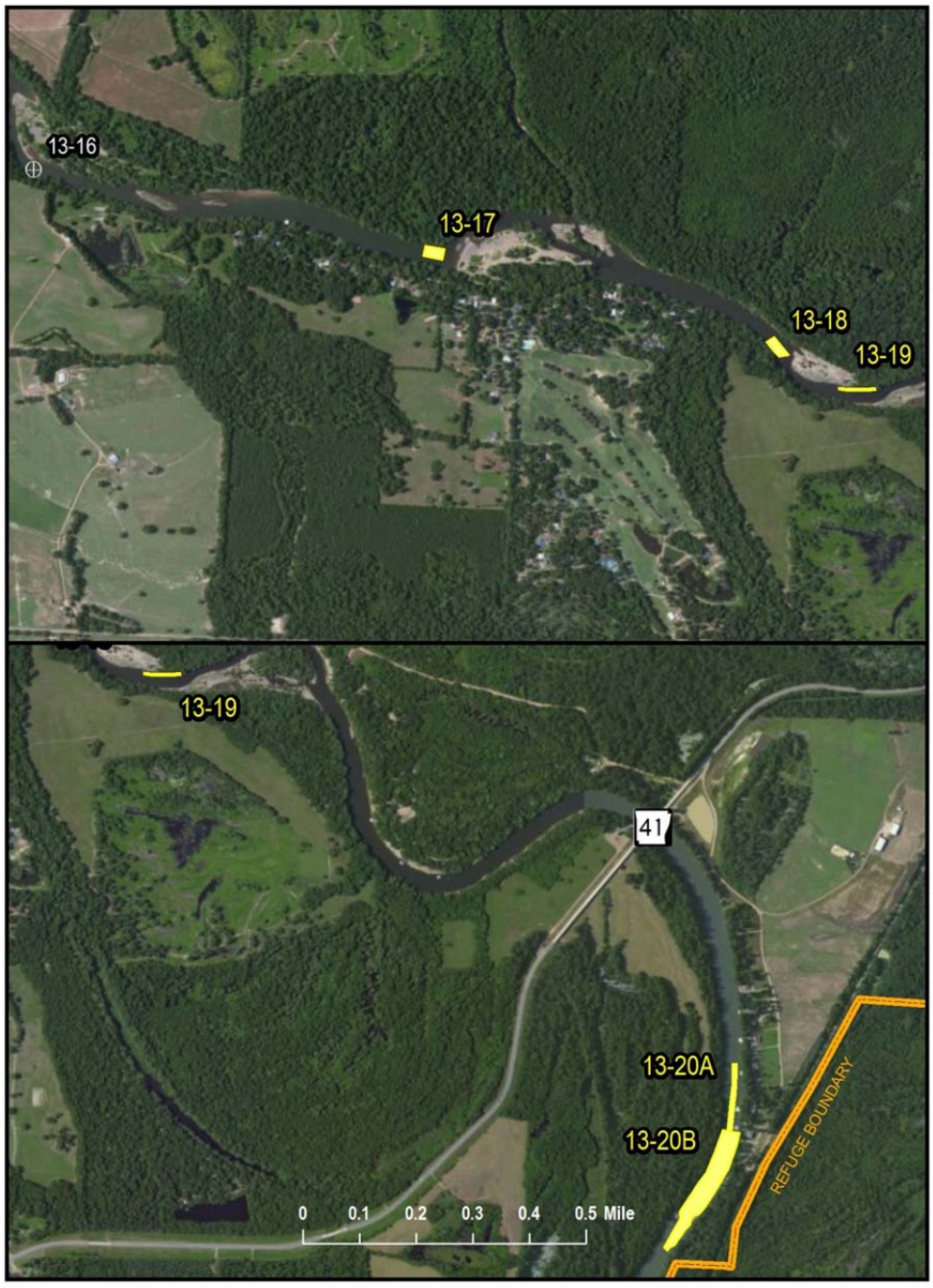


Figure 2c. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

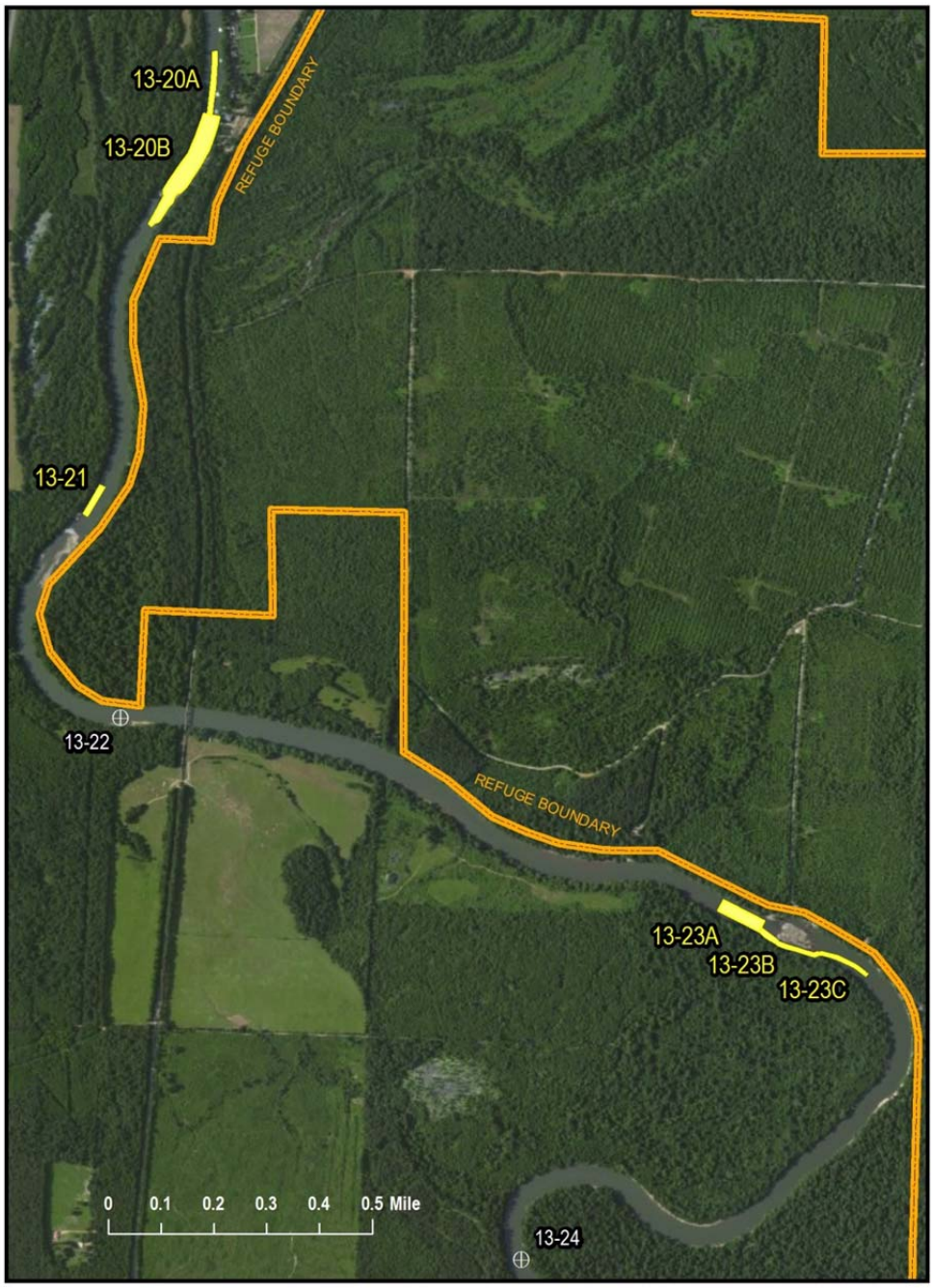


Figure 2d. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

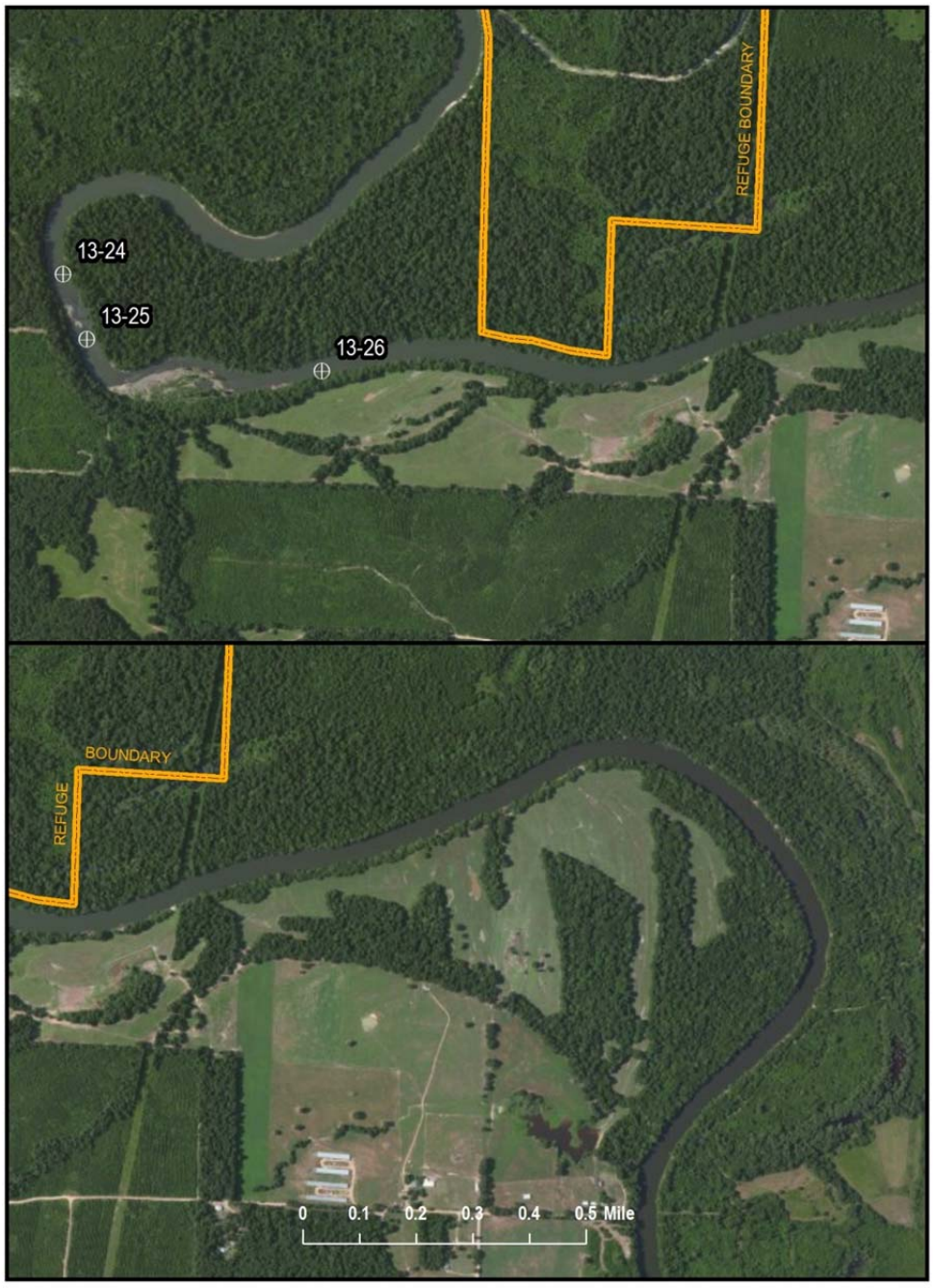


Figure 2e. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

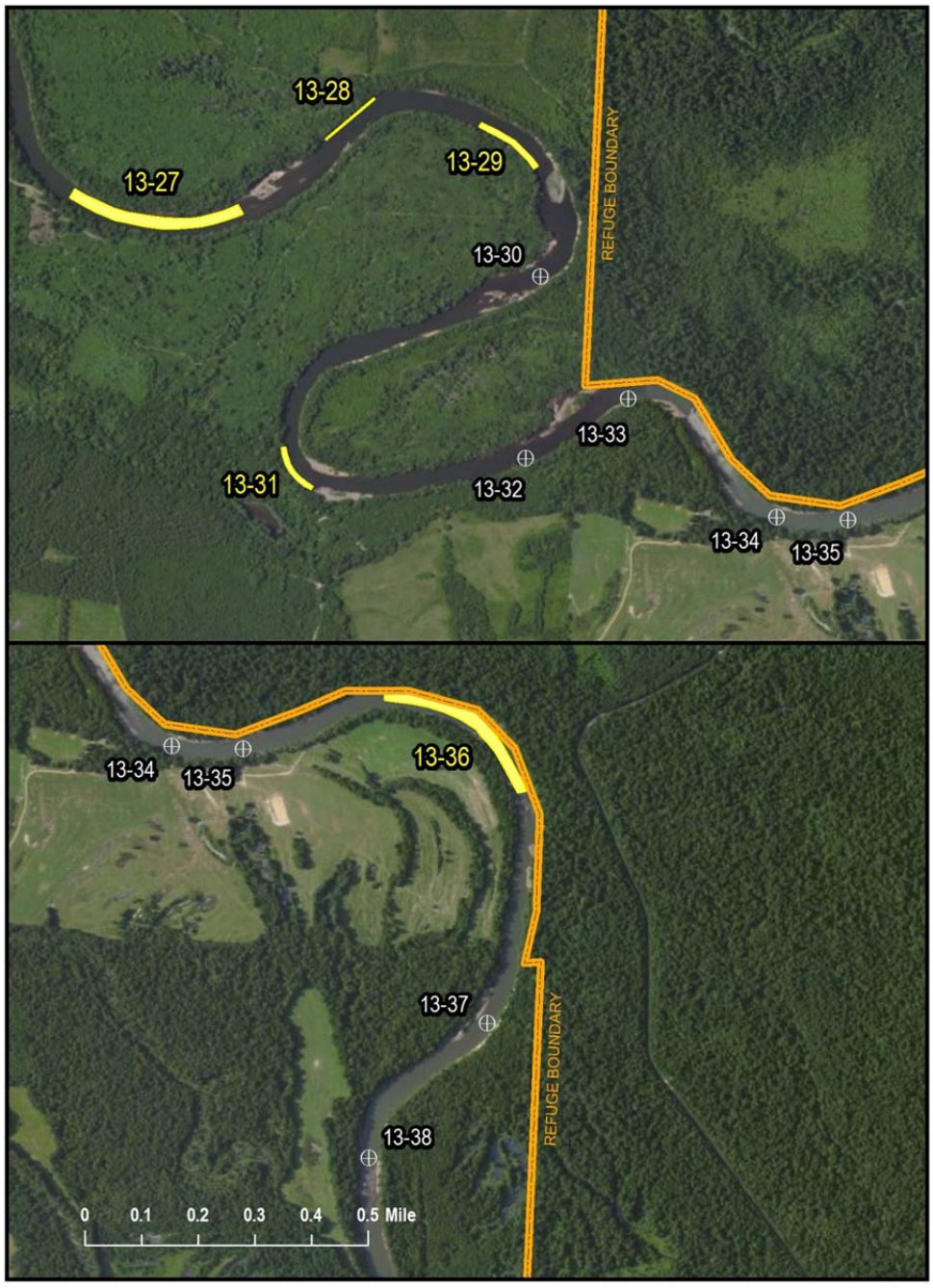


Figure 2f. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

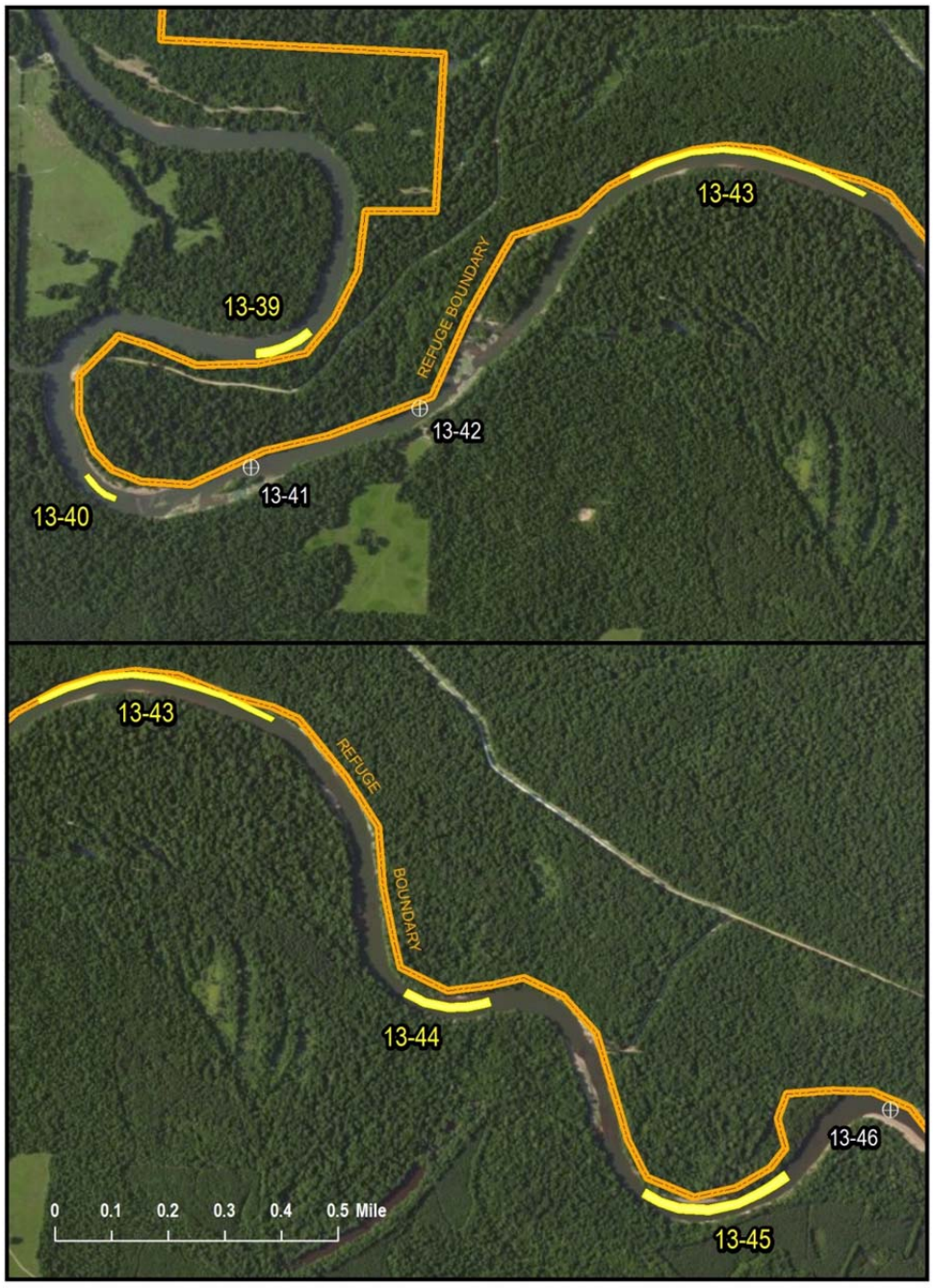


Figure 2g. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

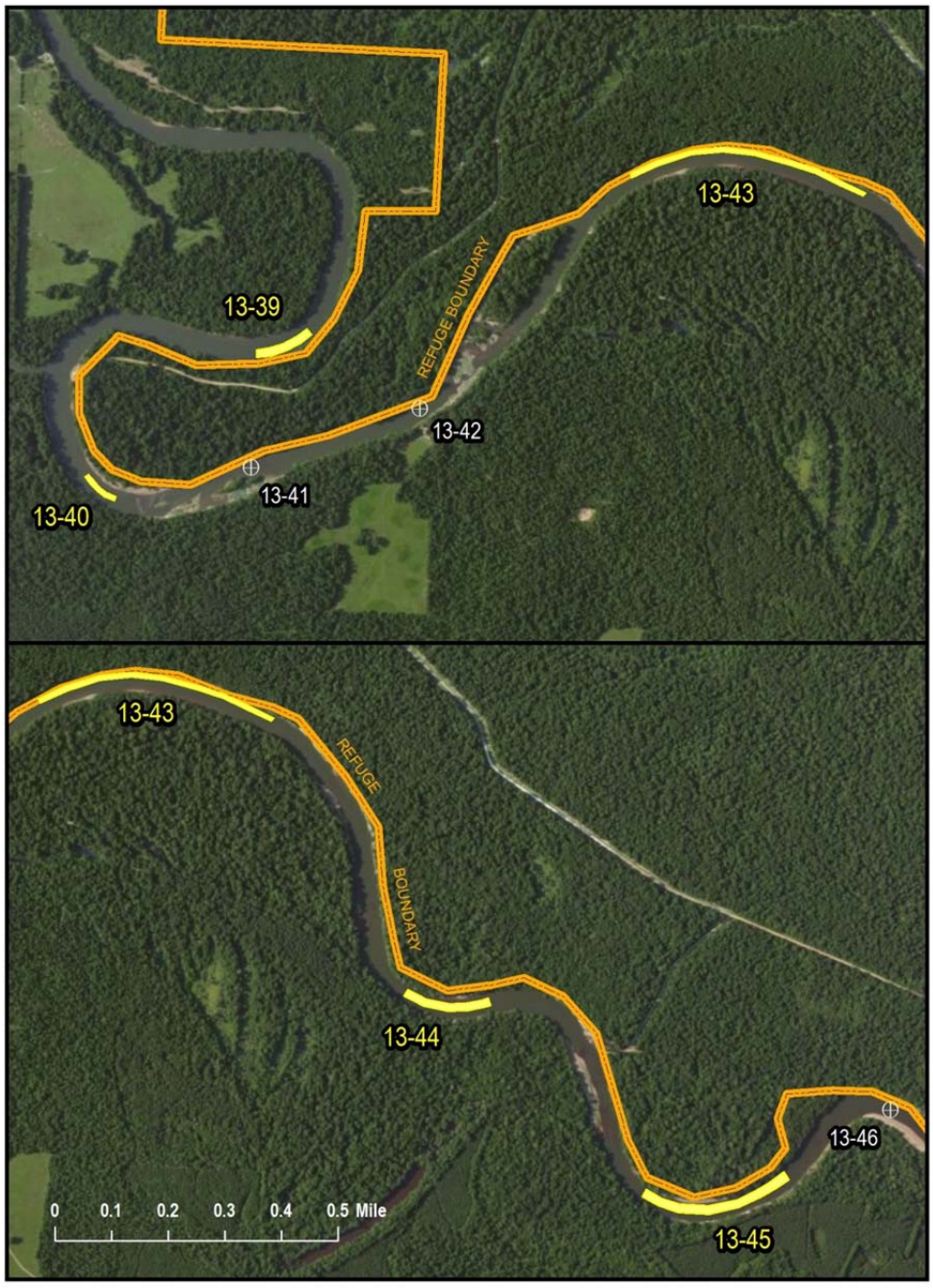


Figure 2h. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

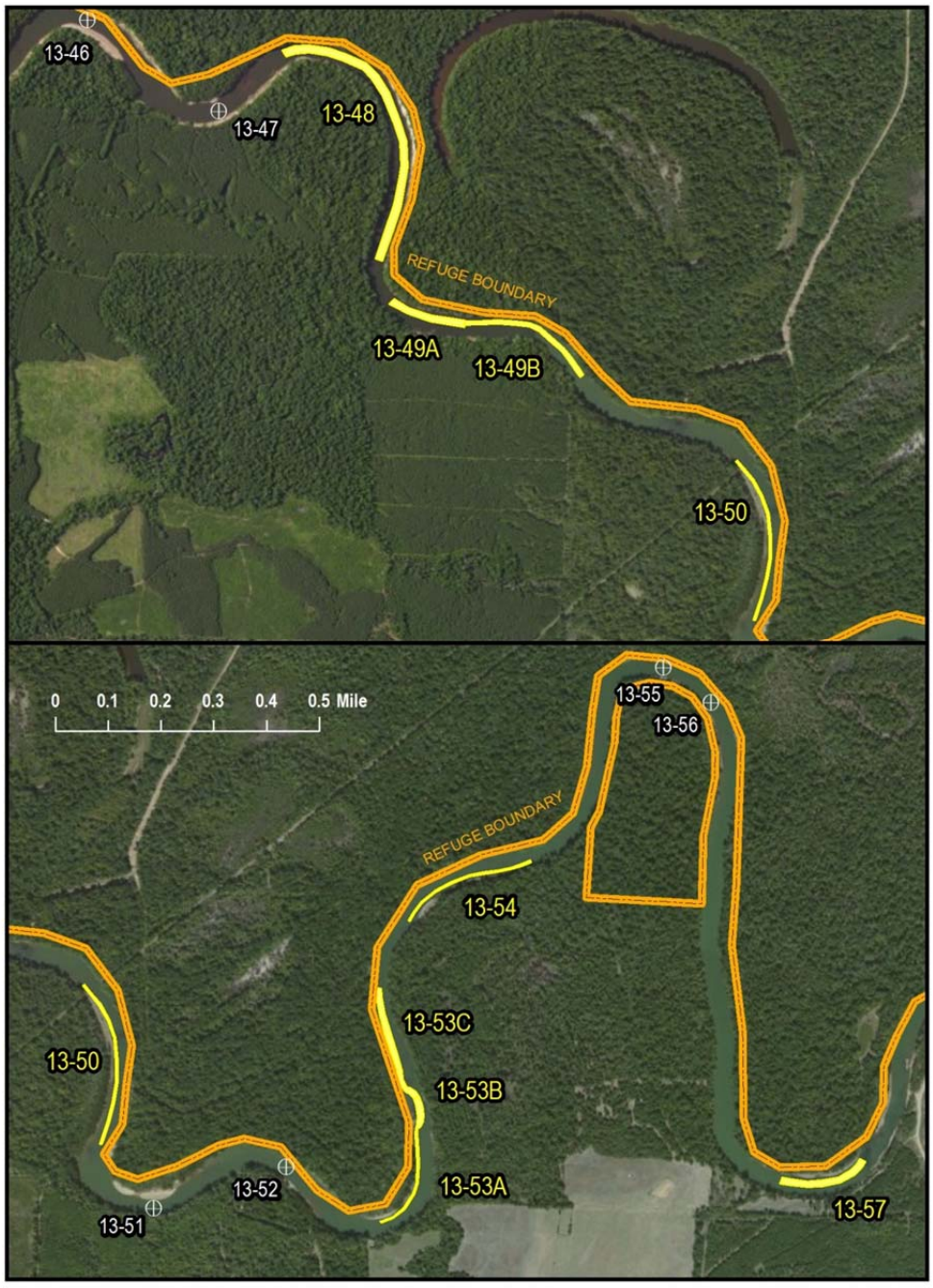


Figure 2i. Map showing sample locations and delineated mussel beds in the Little River, Arkansas – Oklahoma state line to Millwood Lake.

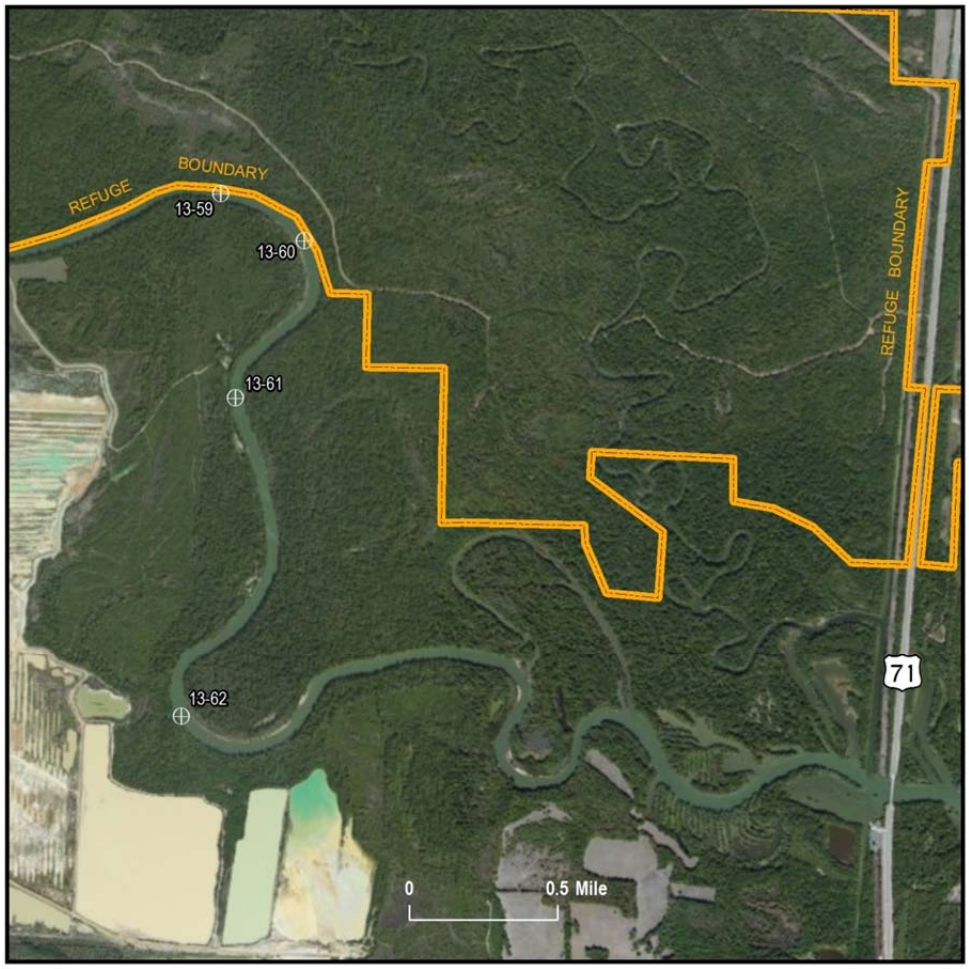


Table 3. Little River, Arkansas – Oklahoma state line to Millwood Lake, mussel bed location, approximate density, and area.

Site Name	Map Label	Start		End		Density (mussels/m^2)	Bed area (m^2)
		Latitude	Longitude	Latitude	Longitude		
LR 09091303	13-03	33.93918	-94.47459	33.93951	-94.47315	10-20	2320
LR 09091307[1]	13-07	33.94117	-94.46233	NR	NR	10-20	240
LR 09091308	13-08	33.93490	-94.45818	33.93463	-94.45775	5-25	2240
LR 09101304	13-12	33.92895	-94.44717	33.92958	-94.44569	20-40	3000
LR 09101306-stratum 1	13-14A	33.93041	-94.43978	33.93038	-94.44097	15-30	4600
LR 09101306-stratum 2	13-14B	33.93038	-94.44097	33.93065	-94.43872	15-30	1875
LR 09101307	13-15	33.93137	-94.42947	33.93070	-94.42861	20-30	2250
LR 09111301-stratum1	13-20A	33.91321	-94.38457	33.91162	-94.38466	30-50	2280
LR 09111301-stratum2	13-20B	33.91162	-94.38466	33.90845	-94.38661	30-50	10000
LR 09111302	13-21	33.90123	-94.38792	33.90055	-94.38890	15-30	1500
LR 09111304 stratum 1	13-23A	33.88988	-94.36750	33.88938	-94.36626	NR	4900
LR 09111304 stratum 2	13-23B	33.88938	-94.36626	33.88871	-94.36444	NR	1300
LR 09111304 stratum 3	13-23C	33.88871	-94.36444	33.88806	-94.36263	NR	1080
LR 09121302	13-28	33.86252	-94.32603	33.86344	-94.32494	20-30	720
LR 09121303	13-29	33.86281	-94.32163	33.86177	-94.31990	20-40	2700
LR 09121305	13-31	33.85450	-94.32745	33.85352	-94.32664	20-50	1920
LR 09131301	13-17	33.92646	-94.41564	33.92648	-94.41500	30-50	1800
LR 09131302	13-18	33.92412	-94.40460	33.92383	-94.40450	25-40	1625
LR 09131303	13-19	33.92297	-94.40281	33.92299	-94.40197	20-30	770
LR 09231304	13-36	33.85428	-94.30586	33.85178	-94.30172	20-40	14400
LR 09241301	13-39	33.82551	-94.29681	33.82498	-94.29843	10-20	2600
LR 09241302	13-40	33.82174	-94.30362	33.82122	-94.30270	5-25	2100
LR 09241305	13-43	33.82965	-94.28702	33.82907	-94.27991	20-30	6600
LR 09241306	13-44	33.82216	-94.27566	33.82197	-94.27303	30-40	4800
LR 09241307	13-45	33.81724	-94.26824	33.81780	-94.26297	40-80	11875
LR 09251303	13-48	33.81855	-94.25420	33.81295	-94.25107	40-60	14800

Table 3. (continued) Little River, Arkansas – Oklahoma state line to Millwood Lake, mussel bed location, approximate density, and area.

Site Name	Map Label	Start		End		Density (mussels/m^2)	Bed area (m^2)
		Latitude	Longitude	Latitude	Longitude		
LR 09251304- stratum 1	13-49A	33.81179	-94.25060	33.81114	-94.24812	20-40	6160
LR 09251304- stratum 2	13-49B	33.81114	-94.24812	33.80975	-94.24432	20-40	3600
LR 09261301	13-50	33.80742	-94.23877	33.80292	-94.23873	15-40	2340
LR 09261304- stratum 1	13-53A	33.80107	-94.22925	33.80373	-94.22806	30-40	2130
LR 09261304- stratum 2	13-53B	33.80373	-94.22806	33.80453	-94.22812	15-25	1600
LR 09261304- stratum 3	13-53C	33.80453	-94.22812	33.80752	-94.22938	12-20	8400
LR 09261305	13-54	33.80941	-94.22845	33.81107	-94.22447	20-30	2670
LR 09261308	13-57	33.80227	-94.21610	33.80297	-94.21349	15-25	4600
LITT 073108POS01	13-27	NR	NR	33.86074	-94.32953	29[2]	15525

[1]LR09091307 coordinates at bed center.
[2]Represents average based on 37 1-m^2 quadrat samples taken in 2008.
NR = not recorded

Table 4. Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey site locations.

Site	Map Label	Latitude	Longitude
LR 09091301	13-01	33.939560	-94.47735
LR 09091302	13-02	33.938860	-94.47605
LR 09091304	13-04	33.941230	-94.46995
LR 09091305	13-05	33.943190	-94.46808
LR 09091306	13-06	33.942260	-94.46302
LR 09101301	13-09	33.932730	-94.45483
LR 09101302	13-10	33.930570	-94.45062
LR 09101303	13-11	33.929040	-94.44899
LR 09101305	13-13	33.930600	-94.44223
LR 09101308	13-16	33.928360	-94.42767
LR 09111303	13-22	33.894860	-94.38732
LR 09111305	13-24	33.880110	-94.37387
LR 09111306	13-25	33.878440	-94.37312
LR 09121301	13-26	33.877710	-94.36588
LR 09121304	13-30	33.858970	-94.31971
LR 09121306	13-32	33.854320	-94.32008
LR 09231301	13-33	33.855870	-94.31696
LR 09231302	13-34	33.852890	-94.31236
LR 09231303	13-35	33.852840	-94.31017
LR 09231305	13-37	33.845870	-94.30257
LR 09231306	13-38	33.842380	-94.30614
LR 09241303	13-41	33.822030	-94.29855
LR 09241304	13-42	33.823590	-94.29338
LR 09251301	13-46	33.819410	-94.26070
LR 09251302	13-47	33.816950	-94.25632
LR 09261302	13-51	33.801390	-94.23672
LR 09261303	13-52	33.802560	-94.23238
LR 09261306	13-55	33.816440	-94.22015
LR 09261307	13-56	33.815500	-94.21857
LR 09271301	13-58	33.807780	-94.20989
LR 09271302	13-59	33.812280	-94.18826
LR 09271303	13-60	33.810020	-94.18334
LR 09271304	13-61	33.802300	-94.18727
LR 09271305	13-62	33.786690	-94.19022

Table 5. Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09091301	LR09091302	LR09091303	LR09091304	LR09091305	LR09091306	LR09091307	LR09091308	LR09101301	LR09101302
Map Label	**13-01**	**13-02**	**13-03**	**13-04**	**13-05**	**13-06**	**13-07**	**13-08**	**13-09**	**13-10**
Actinonaias ligamentina			6 (1)	(1)	1	1	3	50	2	2
Amblema plicata		(1)	1			1	1	13		
Arkansas wheeleri										
Ellipsaria lineolata							1	1		1
Elliptio dilatata			(1)					1		
Fusconaia ebena										
Fusconaia flava										
Lampsilis cardium			1	1				1		(1)
Lampsilis teres			(1)							
Lampsilis sp. B cf. *hydiana*										
Lasmigona costata										
Lasmigona complanata										
Leptodea fragilis				(1)		(1)	1(1)			(1)
Megalonaias nervosa			2				6			
Obliquaria reflexa	1		3	(1)		(2)	2	3	(1)	(1)
Obovaria jacksoniana										
Plectomerus dombeyanus										
Pleurobema ridellii						(1)				
Pleurobema rubrum										
Pleurobema sintoxia							1	2 (2)		(1)
Potamilus purpuratus			(5)			(1)	9	4		1 (2)
Ptychobranchus occidentalis										
Quadrula apiculata										
Quadrula cylindrica cylindrica			(1)					7 (4)		(1)
Quadrula fragosa										
Quadrula metanevra										
Quadrula nobilis										
Quadrula pustulosa	(1)	1	20			9	26	35	6	
Quadrula verrucosa			(3)				5	7	(2)	1
Strophitus undulatus			2							
Truncilla donaciformis										
Truncilla truncata	(1)	1 (1)	10 (2)	2	1	1	9	3	1 (1)	15 (3)
Total Live (dead) Individuals	1 (2)	2 (2)	45 (14)	3 (3)	2 (0)	12 (5)	64 (1)	127 (7)	9 (4)	20 (10)
Search Effort (min)	14	12	36	16	16	18	64	32	28	46
CPUE (# live individuals/min)	0.07	0.17	1.25	0.19	0.13	0.67	1.00	3.97	0.32	0.43
Species Richness	3	3	13	5	2	8	11	13	5	10

Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09101303	LR09101304	LR09101305	LR09101306	LR09101307	LR09101308	LR09111301	LR09111302	LR09111303	LR09111304
Map Label	**13-11**	**13-12**	**13-13**	**13-14**	**13-15**	**13-16**	**13-20**	**13-21**	**13-22**	**13-23**
Actinonaias ligamentina	2	9		100	11		12	7	3	27
Amblema plicata	2	2			1		3	1	2	6
Arkansia wheeleri										
Ellipsaria lineolata		1			1		2	1		
Elliptio dilatata		1								
Fusconaia ebena										
Fusconaia flava		2		1	1					2
Lampsilis cardium	3	4		3						7
Lampsilis teres				(3)						
Lampsilis sp. B cf. *hydiana*										
Lasmigona costata	(1)									
Lasmigona complanata		1								
Leptodea fragilis				1	(2)			7 (1)		(1)
Megalonaias nervosa	1			100	1		1		1	2
Obliquaria reflexa	3	13		9	6			9		10
Obovaria jacksoniana				1	2		24	2	9	1
Plectomerus dombeyanus	1	1		6			3	1		1
Pleurobema ridellii							8	2	2	
Pleurobema rubrum					(1)		3			
Pleurobema sintoxia	1	1		25	3		2	1	1	3
Potamilus purpuratus	2	1		1			1	3	3	1
Ptychobranchus occidentalis							(2)			(1)
Quadrula apiculata					2 (1)		2			
Quadrula cylindrica cylindrica		3		3	(1)		5	2 (1)	1	(2)
Quadrula fragosa							(1)			
Quadrula metanevra		2								(2)
Quadrula nobilis										
Quadrula pustulosa	4	93		100	24		42	29	38	44
Quadrula verrucosa	1	14		3	6		3	1	5	
Strophitus undulatus	2			1			1			1
Truncilla donaciformis		2		2	(1)		1			
Truncilla truncata	7	45		18	13		23	11	2	13
Total Live (dead) Individuals	29 (1)	195	0	374 (3)	71 (6)	0	136 (3)	77 (2)	67	118 (7)
Search Effort (min)	40	80	10	94	106	30	40	34	60	52
CPUE (# live individuals/min)	0.73	2.44	0.00	3.98	0.67	0.00	3.40	2.26	1.12	2.27
Species Richness	13	17	0	17	16	0	19	14	11	18

Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09111305	LR09111306	LR09121301	LR09121302	LR09121303	LR09121304	LR09121305	LR09121306	LR09131301	LR09131302
Map Label	**13-24**	**13-25**	**13-26**	**13-28**	**13-29**	**13-30**	**13-31**	**13-32**	**13-17**	**13-18**
Actinonaias ligamentina		2	2	4	16		16	7	11	40
Amblema plicata		3	1	3 (1)	5		3	1	3	30
Arkansas wheeleri										1
Ellipsaria lineolata			1		1 (1)	2	1			
Elliptio dilatata										
Fusconaia ebena										
Fusconaia flava			3		1				1	2
Lampsilis cardium					5		1		2	1
Lampsilis teres			(1)					1	1	(1)
Lampsilis sp. B cf. *hydiana*										
Lasmigona costata										(1)
Lasmigona complanata										
Leptodea fragilis			11	(1)	(1)		(3)			
Megalonaias nervosa		2	1		2				6	4
Obliquaria reflexa		3	12	11	13		16	18	1	10
Obovaria jacksoniana			1	1	1				1	1
Plectomerus dombeyanus				1	1		3			(1)
Pleurobema ridellii									1	
Pleurobema rubrum			(1)	(1)	2					
Pleurobema sintoxia			2	1			3	1	5	
Potamilus purpuratus		1		(2)	2		7	1	1	3
Ptychobranchus occidentalis					1					1
Quadrula apiculata				(1)		(1)			1	
Quadrula cylindrica cylindrica			(2)	4 (4)	2 (1)		2	(2)	1 (1)	2 (1)
Quadrula fragosa										
Quadrula metanevra				1					(1)	
Quadrula nobilis							(1)		1	
Quadrula pustulosa		6	25	72	53	4	41	22	52	36
Quadrula verrucosa			1	2			3		2	3 (2)
Strophitus undulatus									1	1
Truncilla donaciformis				1 (1)						
Truncilla truncata		1	10	11	5	2	16	3	25	9
Total Live (dead) Individuals	0	18	70 (4)	112 (11)	110 (3)	8 (1)	112 (4)	54 (2)	116 (2)	144 (6)
Search Effort (min)	20	32	76	62	40	20	60	50	56	62
CPUE (# live individuals/min)	0.00	0.56	0.92	1.81	2.75	0.40	1.87	1.08	2.07	2.32
Species Richness	0	7	15	16	16	4	14	9	19	18

Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09131303	LR09231301	LR09231302	LR09231303	LR09231304	LR09231305	LR09231306	LR09241301	LR09241302	LR09241303
Map Label	**13-19**	**13-33**	**13-34**	**13-35**	**13-36**	**13-7**	**13-38**	**13-39**	**13-40**	**13-41**
Actinonaias ligamentina	83	4	1	2	27	24	1	8	28	1
Amblema plicata	2				5			3	2	
Arkansas wheeleri								(1)		
Ellipsaria lineolata	1				1	1		1	(1)	
Elliptio dilatata										
Fusconaia ebena									1	
Fusconaia flava					2	3			1	
Lampsilis cardium	1			1	2	1				
Lampsilis teres										
Lampsilis sp. B cf. *hydiana*										
Lasmigona costata	1									
Lasmigona complanata										
Leptodea fragilis			(1)		(1)					
Megalonaias nervosa									1	
Obliquaria reflexa	3		3	4	11	11	2	19	26	2
Obovaria jacksoniana		1			1			1		
Plectomerus dombeyanus						1				
Pleurobema ridellii					2	1			3	
Pleurobema rubrum					1					
Pleurobema sintoxia	2									
Potamilus purpuratus			(1)		2			2	(1)	
Ptychobranchus occidentalis	(2)									
Quadrula apiculata										
Quadrula cylindrica cylindrica	5 (3)			(1)	1 (3)			1		
Quadrula fragosa										
Quadrula metanevra	1				1 (1)					
Quadrula nobilis					(1)					
Quadrula pustulosa	39	3	5	11	68 (1)	28		39	49	1
Quadrula verrucosa		(1)			4	3	1	1	1	
Strophitus undulatus										
Truncilla donaciformis										
Truncilla truncata	5	2			7	8		8	4	2
Total Live (dead) Individuals	143 (5)	10 (1)	9 (2)	18 (1)	135 (7)	81	4	83 (1)	116 (2)	6
Search Effort (min)	54	68	24	28	64	40	14	48	72	20
CPUE (# live individuals/min)	2.65	0.15	0.38	0.64	2.11	2.03	0.29	1.73	1.61	6.00
Species Richness	12	5	5	5	17	10	3	11	12	4

Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09241304	LR09241305	LR09241306	LR09241307	LR09251301	LR09251302	LR09251303	LR09251304	LR09261301	LR09261302
Map Label	**13-42**	**13-43**	**13-45**	**13-46**	**13-47**	**13-47**	**13-48**	**13-49**	**13-50**	**13-51**
Actinonaias ligamentina		20	20	20	6	6	32	22	9	1
Amblema plicata	(1)	1	7 (1)	3	1	2	11	8	2	
Arkansas wheeleri										
Ellipsaria lineolata		3	19			1	(1)	3	1	2
Elliptio dilatata			1							
Fusconaia ebena			1						(1)	
Fusconaia flava	1		4	2		1		1		
Lampsilis cardium		3	4 (1)	1	(1)	1 (2)	2	5	3	
Lampsilis teres								(1)		
Lampsilis sp. B cf. *hydiana*	(1)									
Lasmigona costata				2			1			
Lasmigona complanata										
Leptodea fragilis	(1)	(1)	(2)		(1)	(2)	(1)		1 (1)	
Megalonaias nervosa		3	1				2	2		
Obliquaria reflexa	1	7	17	11	2 (2)	2 (1)	20	12	12	2
Obovaria jacksoniana		3	3 (1)	(1)	(1)		2	1	1	
Plectomerus dombeyanus	1		2				2	4		
Pleurobema ridellii			1		1 (1)	(1)	2	2	1	
Pleurobema rubrum				1			1		1	
Pleurobema sintoxia			1	4						
Potamilus purpuratus	1	3	2	2	(1)		3	1 (1)	6	2
Ptychobranchus occidentalis		1	9							
Quadrula apiculata										
Quadrula cylindrica cylindrica			(8)	7 (20)	(2)	2 (1)		1	1	
Quadrula fragosa										
Quadrula metanevra										
Quadrula nobilis			(1)				2			1
Quadrula pustulosa	5	58	55	68	10 (1)	29 (1)	52	69	37	8
Quadrula verrucosa		3	9		(1)		5	2	1	
Strophitus undulatus						1				
Truncilla donaciformis			1		(3)	(1)				
Truncilla truncata		5	3	3		5	6	9 (1)	5	3 (1)
Total Live (dead) Individuals	9 (3)	110 (1)	160 (14)	124 (21)	20 (14)	50 (9)	143 (2)	142 (3)	81 (2)	19 (1)
Search Effort (min)	24	50	46	40	28	48	56	64	72	30
CPUE (# live individuals/min)	0.38	2.20	3.48	3.10	0.71	1.04	2.55	2.22	1.13	0.63
Species Richness	8	13	22	12	12	13	16	16	15	7

Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name									
	LR09261303	**LR09261304**	**LR09261305**	**LR09261306**	**LR09261307**	**LR09261308**	**LR09271301**	**LR09271302**	**LR09271303**	**LR09271304**
Map Label	13-52	13-53	13-54	13-55	13-56	13-57	13-58	13-59	13-60	13-61
Actinonaias ligamentina	1	73	43	6		31		8	1	
Amblema plicata		4	1	1		1				
Arkansas wheeleri										
Ellipsaria lineolata	(1)	5		1				8		
Elliptio dilatata										
Fusconaia ebena										
Fusconaia flava		2								
Lampsilis cardium	1	4	2			5		1	1	
Lampsilis teres										
Lampsilis sp. B cf. *hydiana*										
Lasmigona costata			(1)							
Lasmigona complanata										
Leptodea fragilis										
Megalonaias nervosa		1	1					2		
Obliquaria reflexa		23	4	10		4		12		1
Obovaria jacksoniana		1	1							
Plectomerus dombeyanus			1					1		
Pleurobema ridellii		4	2	1		1		1	1	
Pleurobema rubrum		1								
Pleurobema sintoxia		2								
Potamilus purpuratus	2	1	3			1 (1)		1		
Ptychobranchus occidentalis			1 (1)	(1)		1				
Quadrula apiculata										
Quadrula cylindrica cylindrica		4 (1)	2	1		3 (4)				
Quadrula fragosa								1		
Quadrula metanevra										
Quadrula nobilis								1 (1)		
Quadrula pustulosa	11	66	34	21		32		29	2	3
Quadrula verrucosa	1	3	4					1	2	
Strophitus undulatus										
Truncilla donaciformis										
Truncilla truncata	1	4	4	2		2		8	1	1
Total Live (dead) Individuals	17 (1)	198 (1)	103 (2)	43 (1)	0	81 (5)	0	74 (1)	8	5
Search Effort (min)	26	174	60	34	18	42	26	56	11	9
CPUE (# live individuals/min)	0.65	1.14	0.32	1.26	0.00	1.93	0.00	1.32	0.73	0.56
Species Richness	7	16	15	9	0	10	0	13	6	3

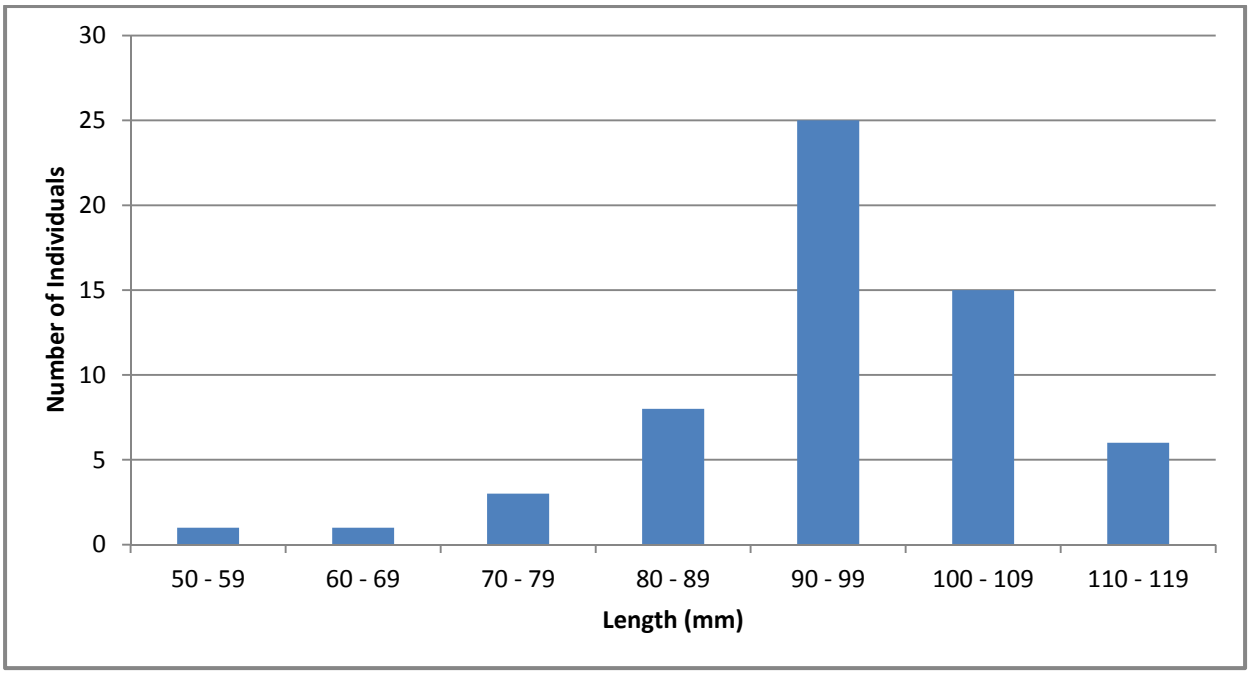
Table 5. (continued) Results of Little River, Arkansas – Oklahoma state line to Millwood Lake, qualitative mussel survey conducted in September, 2013. Live individuals (dead individuals). CPUE = catch per unit effort (live individuals/minute).

Species (dead)	Site Name		Total Live Individuals	Total Dead Individuals	Relative Abundance (Live Individuals)
	LR09271305				
Map Label	**13-62**				
Actinonaias ligamentina			812	2	19.82
Amblema plicata			136	4	3.32
Arkansas wheeleri			1	1	0.02
Ellipsaria lineolata			60	4	1.46
Elliptio dilatata			3	1	0.07
Fusconaia ebena			2	1	0.05
Fusconaia flava			30	0	0.73
Lampsilis cardium	1		68	5	1.66
Lampsilis teres			2	7	0.05
Lampsilis sp. B cf. *hydiana*			0	1	0.00
Lasmigona costata			4	3	0.10
Lasmigona complanata			1	0	0.02
Leptodea fragilis			21	24	0.51
Megalonaias nervosa			142	0	3.47
Obliquaria reflexa	1		365	8	8.91
Obovaria jacksoniana			59	2	1.44
Plectomerus dombeyanus	1		31	1	0.76
Pleurobema ridellii			36	3	0.88
Pleurobema rubrum			10	3	0.24
Pleurobema sintoxia			61	3	1.49
Potamilus purpuratus			73	15	1.78
Ptychobranchus occidentalis	1		15	7	0.37
Quadrula apiculata			5	3	0.12
Quadrula cylindrica cylindrica			60	64	1.46
Quadrula fragosa			1	1	0.02
Quadrula metanevra			5	4	0.12
Quadrula nobilis			5	5	0.12
Quadrula pustulosa	5		1619	4	39.52
Quadrula verrucosa			98	9	2.39
Strophitus undulatus			10	0	0.24
Truncilla donaciformis			7	6	0.17
Truncilla truncata			355	10	8.66
Total Live (dead) Individuals	9		4097	201	100.00
Search Effort (min)	10		-	-	-
CPUE (# live individuals/min)	0.90		-	-	-
Species Richness	5		-	-	-

Table 6. Results of quantitative survey of known Ouachita rock pocketbook (*Arkansia wheeleri*) sites in the Little River, Arkansas – Oklahoma state line to Millwood Lake, 2008.

Species (dead)	Site Name	Relative Abundance	Population Numerical Standing Crop
	LITT073108POS01		
Map Label	**13-27**		
Actinonaias ligamentina	159	14.90	65037 ± 21855
Alasmidonta marginata			
Amblema plicata	10	0.93	4196 ± 2895
Arkansas wheeleri	1	0.01	419 ± 730
Cyprogenia aberti			
Ellipsaria lineolata	11	1.03	4616 ± 2309
Elliptio dilatata			
Fusconaia ebena	1	0.01	419 ± 730
Fusconaia flava	30	2.81	12588 ± 4296
Lampsilis cardium	10	0.93	4196 ± 2895
Lampsilis teres			
Lampsilis sp. B cf. *hydiana*			
Lasmigona costata			
Lasmigona complanata			
Leptodea fragilis	3	0.03	1259 ± 1614
Megalonaias nervosa	9	0.84	
Obliquaria reflexa	114	10.68	
Obovaria jacksoniana	1	0.01	419 ± 730
Plectomerus dombeyanus	4	0.04	
Pleurobema ridellii	11	1.03	4616 ± 2309
Pleurobema rubrum			
Pleurobema sintoxia	2	0.02	839 ± 1460
Potamilus purpuratus			
Ptychobranchus occidentalis	3	0.03	1259 ± 1614
Quadrula apiculata	3	0.03	1259 ± 1229
Quadrula cylindrica cylindrica	5	0.05	2098 ± 1539
Quadrula fragosa			
Quadrula metanevra			
Quadrula nobilis			
Quadrula pustulosa	484	45.36	203084 ± 34868
Quadrula verrucosa	31	2.90	13007 ± 5390
Strophitus undulatus			
Truncilla donaciformis			
Truncilla truncata	169	15.83	70911 ± 13791
Total live Individuals	1067	100.00	-
Mean Density	28.8	-	-
Community Numerical Standing Crop	447707 ± 73066	-	-
Species Richness	20	-	-

Figure 3. Length frequency distribution of rabbitsfoot (*Quadrula cylindrica cylindrica*) from the Little River, Arkansas – Oklahoma state line to Millwood Lake, in 2013.



A total of 1,067 individuals and 20 species were sampled quantitatively from one bed (Site Name LITT073108POS01) in 2008 (Table 6). Four species comprised 87 percent of the total. Species dominance was similar to qualitative surveys for the entire study area (Table 5). Pimpleback (*Quadrula pustulosa*) was the dominant species comprising 45 percent of the total. Mucket (*Actinonaias ligamentina*), Threehorn Wartyback (*Obliquaria reflexa*), and Deertoe (*Truncilla truncata*) comprised the remainder of the 87 percent (14.9, 10.7, and 15.8 percent, respectively). Community numerical standing crop was 447,707 ± 73,066 individuals. One live Ouachita Rock Pocketbook was collected with a population estimate of 419 ± 730.

DISCUSSION

Species Richness and Abundance

Mussels vary considerably with respect to their habitat preferences. Fourteen species previously documented from the Little River were not collected during our survey. Seven of these species prefer sluggish silt (mud) substrate areas in creeks, rivers, and reservoirs (*i.e.*, backwater side channels, pools, mud flats in lakes). It is not surprising these species were absent from our survey given our focus on locating mussel beds in the main channel. Three species, Rock Pocketbook (*Arcidens confragosus*), Pink Heelsplitter (*Potamilus ohiensis*), and Wartyback (*Quadrula nodulata*), are only known from reaches downstream of Millwood Lake. Pink Mucket (*Lampsilis abrupta*) is only known from one record near the Arkansas – Oklahoma state line; if still extant, it likely has always been a rare species in the Little River.

We suspect specimens historically identified as Mapleleaf (*Quadrula quadrula*) are likely Gulf Mapleleaf (*Quadrula nobilis*); as such, no attempt was made to differentiate these closely related and morphologically similar species. Rainbow (*Villosa iris*) is typically found under large cobble and boulders. We did not encounter the preferred habitat type for Rainbow during our survey. One new species occurrence for Little River, Louisiana Pigtoe (*Pleurobema riddellii*), was documented in this survey.

Mussel assemblages typically consist of a few dominant, common species and many more less abundant or rare species (Vaughn 1997). Species dominance has been documented to shift in medium to large rivers (*e.g*., White, Ouachita, Saline, and Buffalo Rivers) from upstream to downstream reaches as physical characteristics (*e.g.,* microhabitat and macrohabitat variables) of the river and host fish presence and abundance change (Christian 1995; Posey 1997; Davidson and Clem 2002, 2004; Matthews *et al.* 2009). We observed a similar community composition shift for species dominance in our study area compared to Vaughn and Taylor (1999) sites located between Pine Creek Reservoir and the Mountain Fork confluence. For example, Threeridge (*Amblema plicata*) is the dominant species at 20 of 23 sites with live mussels upstream of Mountain Fork (Vaughn and Taylor 1999). Among five sites surveyed by Galbraith *et al*. (2005), threeridge and bankclimber (*Plectomerus dombeyanus*) were dominant followed by mucket and pimpleback. Threeridge was never dominant at sites in our study area, representing a shift in species dominance downstream of Mountain Fork. Mussels downstream of reservoirs are likely affected both directly through physical stress (*e.g*., scour, temperature) and indirectly through changes in habitat, food, and fish host availability (Vaughn and Taylor 1999). However, it is not apparent whether this shift in species dominance can be attributed to natural changes, altered flow and temperature regimes resulting from dam operation at Broken Bow Reservoir, and to a lesser extent Pine Creek Reservoir, or a combination of both.

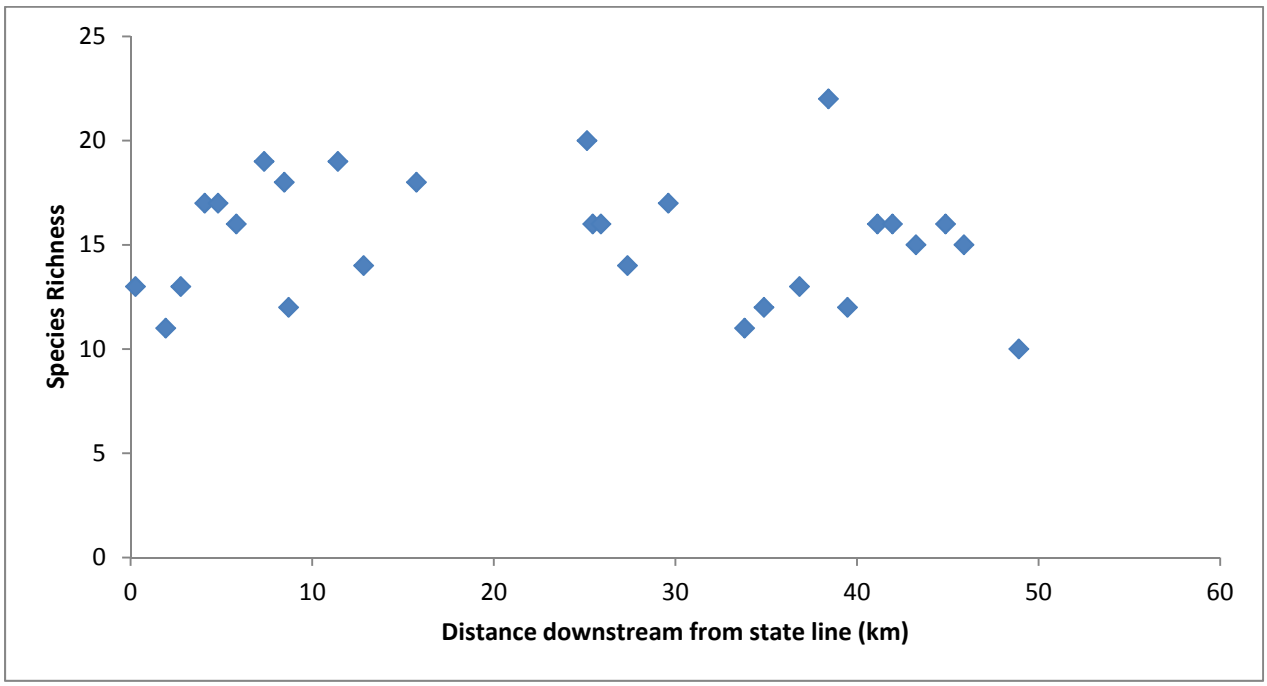
Vaughn and Taylor (1999) documented a mussel extinction gradient downstream of Pine Creek Reservoir and the Mountain Fork confluence (*i.e*., Broken Bow Reservoir inflow) due to altered flow and temperature regimes. There were no apparent trends in species richness and mussel abundance (number of individuals collected per minute) between the state line and Millwood Lake (Figures 4 and 5). Therefore, our data suggest sufficient stream distance occurs between the Mountain Fork confluence and Arkansas – Oklahoma state line for mussel populations to overcome, at least in part, effects of impoundment releases documented by Vaughn and Taylor (1999) between the Mountain Fork confluence and state line. Other anthropogenic factors undoubtedly are influencing the Little River mussel fauna. Vaughn and Taylor (1999) did not find any correspondence between siltation and the linear recovery of mussel abundance and species richness. Further investigation is required to determine factors influencing the Little River mussel fauna downstream of the Arkansas – Oklahoma state line.

Management Considerations and Monitoring Recommendations

Protection or improvements in the status of mussels in the Little River requires proper management of the watershed and cooperative efforts of stakeholders. U.S. Army Corps of Engineers reservoir management at Pine Creek Reservoir, Broken Bow Reservoir, and Millwood Lake has deleterious effects on species richness and abundance. Efforts focused on providing more consistent flows and stable

temperatures, or increased temperatures further downstream of the dams, (*e.g.*, reducing high flows and mixing cold and warm water) should improve habitat conditions for all fish and mussels in the Little River. Implementation of best management practices to reduce sediment runoff from construction sites, county road maintenance, and other land use activities (*e.g.*, forestry) will further improve water quality in the Little River. Execution of these management strategies should improve and/or sustain the quality and quantity of vegetative cover in riparian areas, decrease siltation, and subsequently improve habitat quality for mussels.

Figure 4. Mussel species richness in defined mussel beds versus distance downstream from the Arkansas – Oklahoma state line.

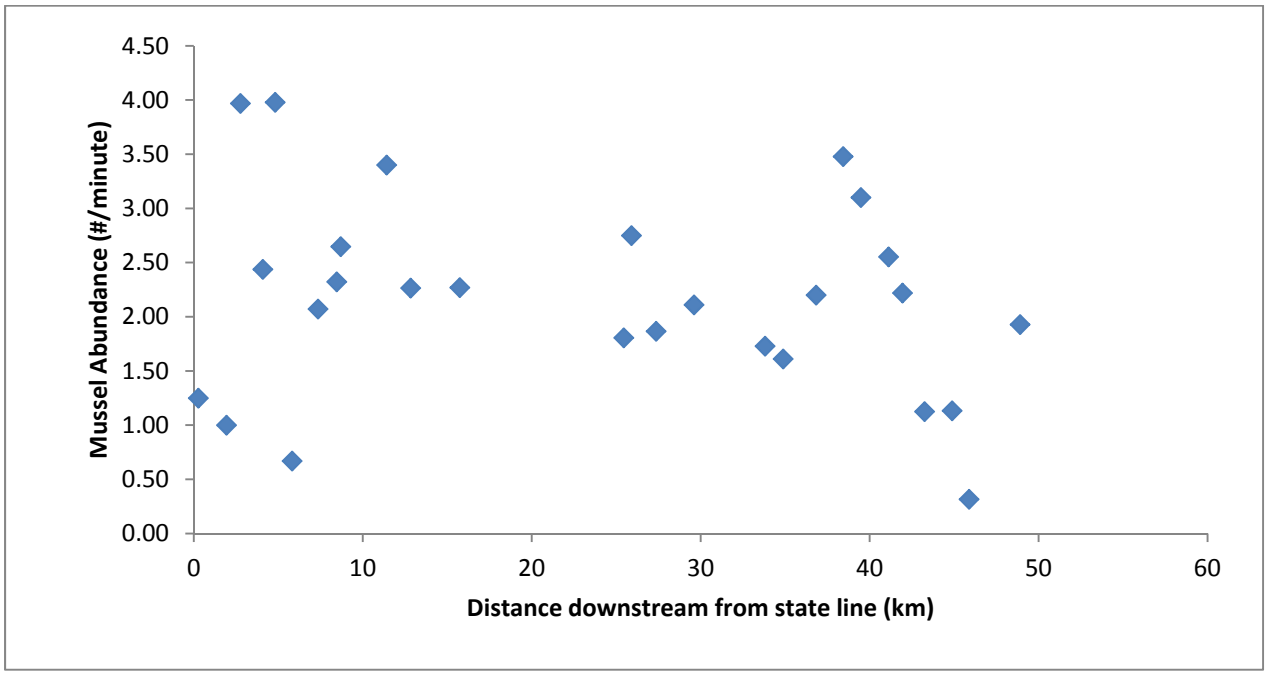


Terrestrial community assessments and monitoring will promote the identification of key areas such as mussel beds and encourage recruitment of landowners or potential partners interested in compatible ecological management options. In order to successfully develop long-term management for rare species, qualitative and quantitative monitoring must be conducted and tailored towards the target mussel species (*i.e.*, threatened and endangered species, "at-risk" species). Information obtained from monitoring activities will establish baseline data that will be used to modify future stewardship activities.

Proper management of native mussel resources includes periodic monitoring. Implementation of a "mussel monitoring program" is needed at Pond Creek National Wildlife Refuge to establish and track population size, demography, recruitment, and status of threatened and endangered mussels (*e.g.*,

Ouachita Rock Pocketbook, Winged Mapleleaf, and Rabbitsfoot), as well as "at-risk" species. Mussel beds with federally protected species should be quantitatively assessed and monitored at a 7 – 10 year interval. Mussel beds identified in this study as containing preferred habitat for Ouachita Rock Pocketbook should be prioritized and targeted for a more intensive sample effort. Identification of sites inhabited by Ouachita Rock Pocketbook are necessary to support a propagation and augmentation program.

Figure 5. Mussel abundance (number/minute) in defined mussel beds versus distance downstream from the Arkansas – Oklahoma state line.



ACKNOWLEDGEMENTS

We thank R. Eastridge, J. Ertel, R. McKay, J. Edwardson, B. Posey, K. Irwin, and M. Wine for field assistance. This project was funded by the Service's Southeast Region Inventory and Monitoring Network in partnership with Pond Creek National Wildlife Refuge, Natchitoches National Fish Hatchery and Arkansas Ecological Services Field Office. We extend special thanks to P. Gideon for his gracious hospitality during this project.

LITERATURE CITED

Christian, A.D. 1995. Analysis of the commercial mussel beds in the Cache and White rivers in Arkansas. Unpublished M.S. thesis, Arkansas State University, Jonesboro. 197 pp.

Christian, A.D., and J.L. Harris. 2005. Development and assessment of a sampling design for mussel assemblages in large streams. Am. Midl. Nat. 153:284-292.

Davidson, C.L., and S.A. Clem. 2004. The freshwater mussel (Bivalvia: Unionacea) resources in a selected segment of the Saline River: location, species composition and status of mussel beds. Addendum 2. Unpublished report, Arkansas Tech University, Russellville, for the Arkansas Game and Fish Commission, Little Rock. 23pp.

Davidson, C.L., and S.A. Clem. 2002. The freshwater mussel (Bivalvia: Unionacea) resources in a selected segment of the Saline River: location, species composition and status of mussel beds. Unpublished report, Arkansas Tech University, Russellville, for The Nature Conservancy, Little Rock, Arkansas, and Arkansas Game and Fish Commission, Little Rock. 48 pp.

Ecosearch, Inc. 1987. Status survey of Lampsilis streckeri (Frierson, 1927) and Arcidens wheeleri (Ortmann and Walker, 1912). Unpublished report, U.S. Fish and Wildlife Service, Jackson, MS. 24 pp. + appendices.

Galbraith, H.S., D.E. Spooner, and C.C. Vaughn. 2005. *Arkansia wheeleri* monitoring in the Kiamichi and Little Rivers. Unpublished report submitted to Oklahoma Department of Wildlife Conservation, Oklahoma City, OK. 59 pp.

Haag, W.L. and M.L. Warren. 1997. Host fishes and reproductive biology of six freshwater mussel species from the Mobile Basin, U.S.A. J. N. Am. Benth. Soc. 16:576-585.

Harris, J.L., and M.E. Gordon. 1987. Distribution and status of rare and endangered mussels (Mollusca: Margaritiferidae, Unionidae) in Arkansas. Proceedings Arkansas Academy of Science 41:49-56.

Harris, J.L., P. Rust, S.W. Chordas, III, and G.L. Harp. 1993. Distribution and population structure of freshwater mussels (Unionidae) in Lake Chicot, Arkansas. Proceedings Arkansas Academy of Science 47:38-43.

Matthews. W.J., C.C. Vaughn, K.B. Gido, and E. Marsh-Matthews. 2005. Southern plains rivers. Pages 283-325 in A.C. Benke and C.E. Cushing, editors, Rivers of North America. Elsevier Academic Press, Massachusetts.

Matthews, M., F. Usrey, S.W. Hodges, J.L. Harris, and A.D. Christian. 2009. Species richness, distribution, and relative abundance of freshwater mussels (Bivalvia: Unionidae) of the Buffalo National River, Arkansas. Journal of the Arkansas Academy of Science 63:113-130.

Posey, W.R., Jr. 1997. Location, species composition, and community estimates for mussel beds in the St. Francis and Ouachita rivers in Arkansas. Unpublished M.S. thesis, Arkansas State University, Jonesboro. 178 pp.

Sampford, M.R. 1962. An introduction to sampling theory with applications to agriculture. Oliver and Boyd Ltd., Edinburgh.

Seagraves, J. 2006. The reproductive biology and habitat characteristics of the endangered Ouachita rock pocketbook, *Arkansia wheeleri*, (Ortmann and Walker, 1912). Unpublished M.S. thesis, Arkansas State University, Jonesboro. 64 pp.

Turgeon D.D., J.F. Quinn, Jr, A.E. Bogan, E.V. Coan, F.G. Hochberg, W.G. Lyons, P.M. Mikkelsen, R.J. Neves, C.F.E. Roper, G. Rosenberg, B. Roth, A. Scheltema, F.G. Thompson, M. Vecchione, and J.D. Williams. 1998. Common and scientific names of aquatic invertebrates from the United States and Canada: mollusks, 2nd edition. American Fisheries Society, Special Publication 26, Bethesda, Maryland. 526 p.

URS Corporation. 2007. Mussel survey – Little River near Fulton, Arkansas. Unpublished report American Electric Power. 14 p.

Vaughn, C.C. 1994. Survey for *Arkansia wheeleri* in the Little River. Unpublished report submitted to U.S. Fish and Wildlife Service, Tulsa, OK. 23 pp. + Appendix.

Vaughn, C.C., and C.M. Taylor. 1999. Impoundments and the decline of freshwater mussels: a case study of an extinction gradient. Conservation Biology 13:912-920.

Vaughn. C.C. and C.M. Taylor. 2000. Macroecology of a host-parasite relationship. Ecography 23:11-20.

Vaughn, C.C., C.M. Taylor, K.J. Eberhard, and M. Craig. 1995. Survey for *Arkansia wheeleri* in the Little River in eastern Oklahoma and western Arkansas. Unpublished report submitted to U.S. Fish and Wildlife Service, Tulsa, OK. 54 pp.

Woods A.J., Foti, T.L., Chapman, S.S., Omernik, J.M., Wise, J.A., Murray, E.O., Prior, W.L., Pagan, J.B., Jr., Comstock, J.A., and Radford, M., 2004, Ecoregions of Arkansas (color poster with map, descriptive text, summary tables, and photographs): Reston, Virginia, U.S. Geological Survey (map scale 1:1,000,000).